Nemo Winch V3.0

Geared DC Motor Iteration

Board-Level Port Documentation
Wiring Harness and Assembly Manual

Document Type	Technical wiring harness and assembly manual
Board Version	Nemo Winch V3.0
Motor Architecture	External BTS7960 geared DC motor driver
Primary Input	6S LiPo / 24 V nominal battery input
Logic Voltage	5 V and 3.3 V rails
Control MCU:	Seeed Studio XIAO ESP32C3, SKU 113991054
Encoder	AS5600 magnetic absolute encoder
Servo Supply	Internal 5 V or external BEC selectable

Prepared for Assembly Team, Ground Operators, and Payload Integration Team

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1 System Overview

The Nemo Winch V3.0 board is a UAV payload control and interface board designed for a winch mechanism using a geared DC motor and an external BTS7960 motor driver module.

The board performs the following functions:

- Receives raw battery power from the UAV LiPo battery.
- Generates regulated 5 V logic power using the onboard buck converter.
- Interfaces with an external BTS7960 motor driver for high-current motor control.
- Provides ESP32-C3 based control for motor PWM, clutch servo PWM, encoder reading, telemetry, and safety logic.
- Interfaces with the Pixhawk flight controller through UART telemetry and AUX PWM signals.
- Reads spool position using an AS5600 magnetic rotary encoder over I2C.
- Provides manual operator override through physical switch inputs.
- Supports selectable servo power using either internal 5 V or an external high-current BEC.

In this design, the controller is not a bare ESP32-C3 chip. The board uses a plug-in **Seeed Studio XIAO ESP32C3 module, SKU 113991054**. Therefore, all firmware pin mapping, UART routing, I2C routing, and PWM outputs must be checked against the XIAO ESP32C3 pinout before PCB fabrication and harness assembly.

1.1 High-Level Functional Blocks

Table 1: Main Functional Blocks of Nemo Winch V3.0

Block	Function
Main Power Input	Accepts raw UAV battery voltage through XT60 connector J14.
Buck Regulator	Converts raw battery voltage to regulated 5 V for logic and low-current peripherals.
ESP32-C3 Controller	Main processing unit for PWM generation, encoder reading, telemetry, safety logic, and manual control.
I/O Expander	Provides additional GPIO lines for BTS7960 enables, end-stop switch, and manual switch inputs.
BTS7960 Interface	Sends PWM, enable, and logic power signals to the external high-current H-bridge module.
Pixhawk Interface	Receives mission commands and sends telemetry through UART and AUX PWM inputs.
Encoder Interface	Connects to AS5600 magnetic encoder over I2C for spool position measurement.
Servo Power Selector	Allows the clutch servo to be powered from either internal 5 V or external BEC output.
Manual Override	Provides physical up/down control for ground operators.

2 Electrical Power Architecture

The board has four major power domains:

1. **VBAT_IN:** Raw UAV battery voltage, typically 6S LiPo / 24 V nominal.
2. **+5V:** Regulated logic voltage generated by the onboard TPS54302 buck converter.
3. **+3V3:** Low-voltage logic rail for ESP32-C3, I2C sensors, and low-power digital logic.
4. **+VCC Servo Rail:** Selectable servo power rail derived either from internal 5 V or external BEC.

2.1 Power Flow Summary

Table 2: Power Flow Description

Source	Destination	Purpose
J14 XT60	VBAT_IN Rail	Main battery input to the payload system.
VBAT_IN	TPS54302 Buck Regulator	Generates onboard 5 V logic supply.
VBAT_IN	J2 BTS7960 Power Output	Provides unfused high-current motor driver supply.
VBAT_IN	J11 External BEC Input	Sends raw battery voltage to an external servo BEC.
TPS54302 Output	+5V Logic Rail	Powers ESP32 board, BTS7960 logic, pullups, indicators, and low-current logic.
External BEC Output	J13 EXT_BEC	Receives external 6 V or 7.4 V servo supply.
J12 Jumper	+VCC Servo Rail	Selects whether servo receives internal 5 V or external BEC voltage.

2.2 Important Power Safety Notes

- The BTS7960 motor power line is supplied directly from `VBAT_IN`. It is intentionally separated from the low-current logic 5 V rail.
- The clutch servo must not be powered from the internal 5 V rail if it is a high-torque servo with large stall current.
- For high-torque servos, use the external BEC input/output architecture through J11, J13, and J12.
- All power domains must share a common system ground.
- Reverse-polarity protection and fuse protection should be verified before flight use.

3 Connector Summary

Table 3: Complete Connector Summary for Nemo Winch V3.0

Connector	Name	Function
J14	XT60 Main Power Input	Primary raw battery power input from drone LiPo battery.
J2	BTS7960 Power Output	High-current raw battery power output to BTS7960 motor driver.
J3	BTS7960 Logic Header	Carries 5 V, GND, enable signals, and PWM signals to BTS7960 module.
J1	Pixhawk Telemetry Port	UART telemetry link between ESP32-C3 and Pixhawk flight controller.
J7	Winch Control Input Header	AUX PWM command input from Pixhawk for winch and clutch control.
J9	AS5600 Encoder Header	I2C interface for magnetic spool position encoder.
J10	Clutch Servo Header	Standard 3-pin servo output for clutch actuator.
J4	End-Stop Contact Terminal	Safety input from physical end-stop micro-switch.
J5	Manual Switch Header	Ground crew physical spool up/down override switch.
J11	External BEC Input	Sends raw battery voltage to external high-current BEC.
J13	External BEC Output	Receives regulated external servo supply from BEC.
J12	Servo Power Selection Jumper	Selects whether servo rail receives internal 5 V or external BEC voltage.

4 Main Power Interface

4.1 J14 – XT60 Main Power Input

Connector: XT60-M connector
Function: Primary power entry point for the winch payload system. This connector receives raw unregulated voltage directly from the drone battery.

Table 4: J14 Main Power Input Pinout

Pin	Net Name	Description
1	VBAT_IN	Raw battery positive input. Typically 6S LiPo / 24 V nominal.
2	System GND	Battery negative and common system ground.

4.2 J2 – BTS7960 High-Current Power Output

Connector: 2-pin screw terminal or high-current solder pad
Function: Provides raw, unfused high-current power directly to the BTS7960 motor driver module.

Table 5: J2 BTS7960 Power Output Pinout

Pin	Net Name	Description
1	System GND	Ground connection for BTS7960 motor driver power input.
2	VBAT_IN	Raw battery voltage supplied to BTS7960 motor driver.

4.3 Recommended Wiring for J14 and J2

Table 6: Recommended Wire Gauge for Power Lines

Connection	Recommended Wire	Notes
Battery to J14	14 AWG to 12 AWG silicone wire	Depends on total payload current and cable length.
J2 to BTS7960 Power Input	14 AWG to 12 AWG silicone wire	Must handle motor stall current safely.
Ground Return	Same gauge as positive line	Never use thinner ground return wire.

5 Motor Driver Interface

The Nemo Winch V3.0 board uses an external BTS7960 H-bridge module for the geared DC motor. The custom PCB does not carry full motor phase current through the logic section. Instead, it provides high-current raw battery supply, logic power, enable control, and PWM control signals to the BTS7960.

5.1 J3 – BTS7960 Logic Header

Connector: 8-pin female header socket
Function: Logic control interface between Nemo Winch PCB and the BTS7960 motor driver.

Table 7: J3 BTS7960 Logic Header Pinout

Pin	BTS7960 Signal	Connected Net	Function
1	VCC	+5V	Powers BTS7960 logic input buffer.
2	GND	System GND	Common logic ground.
3	R_EN	Expander P0	Enables right/forward half of the H-bridge.

Pin	BTS7960 Signal	Connected Net	Function
4	L_EN	Expander P1	Enables left/reverse half of the H-bridge.
5	R_PWM	ESP32-C3 D0	Forward PWM speed command.
6	L_PWM	ESP32-C3 D10	Reverse PWM speed command.
7	R_IS	Not Connected	Analog current sense output. Currently unused.
8	L_IS	Not Connected	Analog current sense output. Currently unused.

5.2 Motor Direction Logic

Table 8: BTS7960 Control Logic

Mode	R_EN / L_EN	PWM Signal	Result
Motor Off	LOW / LOW	0% duty	Motor disabled.
Spool Up	HIGH / HIGH	R_PWM active, L_PWM LOW	Motor rotates in forward/retract direction.
Spool Down	HIGH / HIGH	L_PWM active, R_PWM LOW	Motor rotates in reverse/release direction.
Brake / Stop	HIGH / HIGH	Both PWM LOW or firmware braking logic	Motor stops according to BTS7960 behavior and firmware strategy.
Fault / Emergency	LOW / LOW	Both PWM LOW	Driver disabled immediately.

5.3 Firmware Safety Recommendation

The firmware should never drive both `R_PWM` and `L_PWM` at the same time. A software deadband should be implemented when changing direction.

Recommended direction change sequence:

1. Set both PWM outputs to 0%.
2. Wait 100–300 ms for motor current decay.
3. Change direction command.
4. Ramp PWM gradually from 0% to target duty.

6 Flight Controller Interfaces

The board provides two independent interfaces to the Pixhawk flight controller:

1. UART telemetry through J1.
2. Hardware PWM command inputs through J7.

6.1 J1 – Pixhawk Telemetry Port

Connector: 6-pin Pixhawk-style telemetry connector
Function: Provides serial UART communication between ESP32-C3 and Pixhawk. This can be used for MAVLink telemetry, winch status reporting, payload state, fault status, and spool position feedback.

Table 9: J1 Pixhawk Telemetry Port Pinout

Pin	Signal	Function
1	+5V	Optional power tie. Use only if power domain direction and voltage compatibility are verified.
2	ESP32-C3 RX	Receives serial data from Pixhawk TX.

Pin	Signal	Function
3	ESP32-C3 TX	Transmits serial data to Pixhawk RX.
4	CTS / Optional	Optional hardware flow control. Can be left unused if not implemented.
5	RTS / Optional	Optional hardware flow control. Can be left unused if not implemented.
6	System GND	Common ground between Pixhawk and winch board.

6.2 UART Wiring Rule

For UART communication, TX and RX must cross:

Table 10: Pixhawk to ESP32 UART Wiring

Pixhawk Signal	Nemo Winch Signal	Description
Pixhawk TX	ESP32-C3 RX	Pixhawk sends data to ESP32-C3.
Pixhawk RX	ESP32-C3 TX	ESP32-C3 sends data to Pixhawk.
Pixhawk GND	System GND	Required common reference.

6.3 J7 – Winch Control Input Header

Connector: 4-pin input header
Function: Receives direct hardware PWM commands from Pixhawk AUX output channels.

Table 11: J7 Winch Control Input Header Pinout

Pin	Signal	Function
1	Winch Control PWM	Input from Pixhawk AUX 1. Used for spool up/down or speed command.
2	Clutch Signal PWM	Input from Pixhawk AUX 2. Used to command clutch servo position.
3	System GND	Common signal ground.
4	ZERO RESET	Hardware reset input to zero encoder/spool reference position.

6.4 Recommended AUX PWM Interpretation

Table 12: Suggested Pixhawk AUX PWM Command Mapping

PWM Range	Command	Action
900–1200 μs	Spool Down	Motor rotates to release payload.
1400–1600 μs	Neutral	Motor stopped.
1800–2100 μs	Spool Up	Motor rotates to retract payload.
Invalid / No Signal	Failsafe	Motor disabled and clutch held in safe position.

7 Sensor and Actuator Ports

7.1 J9 – AS5600 Encoder Header

Connector: 4-pin I2C sensor header
Function: Connects to the AS5600 magnetic absolute rotary encoder mounted on the winch spool shaft.

Table 13: J9 AS5600 Encoder Header Pinout

Pin	Signal	Function
1	+3.3V	Sensor power supply.
2	SCL	I2C clock line from ESP32-C3.
3	SDA	I2C data line from ESP32-C3.
4	System GND	Sensor ground.

7.2 Encoder Function

The AS5600 encoder is used to measure the angular position of the winch spool. From this, the firmware can estimate:

- Number of spool rotations.
- Cable length released.
- Payload depth or drop distance.
- Zero position after reset.
- End-stop verification.

7.3 Cable Length Calculation

The approximate released cable length can be calculated as:

$$L = N \times \pi D$$

where L is the cable length released, N is the number of spool rotations, and D is the effective spool diameter. If the cable winds in multiple layers, the effective diameter changes and must be calibrated experimentally.

7.4 J10 – Clutch Servo Header

Connector: Standard 3-pin servo header
Function: Drives the clutch servo used to mechanically engage or disengage the winch mechanism.

Table 14: J10 Clutch Servo Header Pinout

Pin	Signal	Function
1	PWM3	Servo PWM signal from ESP32-C3 D2.
2	+VCC	Servo power rail selected by jumper J12.
3	System GND	Servo ground.

7.5 Servo PWM Recommendation

Table 15: Recommended Servo PWM Positions

PWM Pulse Width	Servo Position	Function
1000 μs	Clutch Disengaged	Motor mechanically disconnected from spool.
1500 μs	Neutral / Hold	Safe intermediate position if mechanically supported.
2000 μs	Clutch Engaged	Motor mechanically coupled to spool.

8 Operator and Safety Inputs

8.1 J4 – End-Stop Contact Terminal

Connector: 2-pin contact terminal
Function: Connects to a physical micro-switch used as a retract end-stop safety input.

Table 16: J4 End-Stop Contact Terminal Pinout

Pin	Signal	Function
1	Expander P2	End-stop signal input. Pulled HIGH internally and pulled LOW when switch closes.
2	System GND	Switch ground return.

8.2 End-Stop Logic

Table 17: End-Stop Logic Table

Switch State	Input State	Firmware Action
Open	HIGH	Normal operation allowed.
Closed	LOW	Stop motor immediately and block further retraction.
Disconnected	HIGH or floating	Treat as fault unless pull-up and diagnostics are verified.

8.3 J5 – Manual Switch Header

Connector: 3-pin manual control header
Function: Provides physical override control for ground crew to spool the winch up or down without the radio controller.

Table 18: J5 Manual Switch Header Pinout

Pin	Signal	Function
1	Expander P4	Manual spool up input.
2	System GND	Common switch ground.
3	Expander P3	Manual spool down input.

8.4 Manual Switch Operation

A three-position momentary switch is recommended:

- Center position: OFF / neutral.
- Up position: pulls Expander P7 to ground.
- Down position: pulls Expander P6 to ground.

Table 19: Manual Switch Logic

Input State	Command	Action
P4 LOW, P3 HIGH	Manual Up	Retract cable.
P4 HIGH, P3 LOW	Manual Down	Release cable.
P4 HIGH, P3 HIGH	Neutral	Motor stopped.
P4 LOW, P3 LOW	Invalid	Motor disabled for safety.

9 Servo Power Architecture

The Nemo Winch V3.0 board includes a jumper-selectable servo power system. This prevents high-current servo loads from damaging or overloading the onboard 5 V logic supply.

The system uses three connectors:

- J11 – External BEC input from battery.
- J13 – External BEC regulated output return.
- J12 – Servo power selection jumper.

9.1 J11 – External BEC Input

Connector: 2-pin BEC input connector
Function: Sends raw battery voltage to an external high-current BEC.

Table 20: J11 External BEC Input Pinout

Pin	Signal	Function
1	VBAT_IN	Raw battery voltage to external BEC input.
2	System GND	Ground return for external BEC input.

9.2 J13 – External BEC Output

Connector: 2-pin BEC output connector
Function: Receives regulated high-current servo voltage from the external BEC.

Table 21: J13 External BEC Output Pinout

Pin	Signal	Function
1	EXT_BEC	Regulated external BEC voltage, usually 6 V or 7.4 V.
2	System GND	Ground return from external BEC.

9.3 J12 – Servo Power Selection Jumper

Connector: 3-pin jumper header
Function: Selects the power source for the servo rail `+VCC`.

Table 22: J12 Servo Power Selection Jumper Pinout

Pin	Signal	Function
1	EXT_BEC	External BEC servo supply input.
2	+VCC	Selected servo supply output to J10.
3	+5V	Internal regulated 5 V supply.

9.4 Jumper Configuration

Table 23: J12 Servo Power Selection Configuration

Jumper Position	Servo Supply	Use Case
Pins 2–3 shorted	Internal +5V	Use only for small 5 V servo with low stall current.
Pins 1–2 shorted	External BEC	Use for high-torque servo requiring 6 V or 7.4 V.
No jumper installed	Servo unpowered	Safe state for testing signal without servo motion.
Pins 1–3 shorted	Invalid / Dangerous	Do not install. This shorts external BEC to internal 5 V rail.

Warning: Do not short J12 pin 1 and pin 3. This can connect the external BEC output directly to the onboard 5 V regulator output and may permanently damage the PCB, BEC, or connected electronics.

The following table defines the signal allocation for the **Seeed Studio XIAO ESP32C3, SKU 113991054** module used on the Nemo Winch V3.0 PCB.

10 Seeed Studio XIAO ESP32C3 Signal Allocation

Table 24: Seeed Studio XIAO ESP32C3 Signal Allocation for Nemo Winch V3.0

XIAO ESP32C3 Pin / Signal	Connected Function	Description
D0	BTS7960 R_PWM	Forward motor PWM output.
D10	BTS7960 L_PWM	Reverse motor PWM output.
D3	Clutch Servo PWM	Servo signal output to J10.
SCL	AS5600 SCL	I2C clock for magnetic encoder.
SDA	AS5600 SDA	I2C data for magnetic encoder.
TX	Pixhawk RX	UART transmit from XIAO ESP32C3 to Pixhawk telemetry RX.
RX	Pixhawk TX	UART receive into XIAO ESP32C3 from Pixhawk telemetry TX.
GPIO Interrupt	ZERO RESET	Hardware encoder zero reset input.

11 I/O Expander Signal Allocation

Table 25: I/O Expander Signal Allocation

Expander Pin	Connected Signal	Function
P0	BTS7960 R_EN	Enables forward side of motor driver.
P1	BTS7960 L_EN	Enables reverse side of motor driver.
P2	End-Stop Switch	Detects fully retracted payload condition.
P3	Manual Down Switch	Manual spool down command.
P4	Manual Up Switch	Manual spool up command.

12 Assembly Wiring Procedure

12.1 Step 1 – Main Battery Wiring

1. Connect UAV battery positive to J14 pin 1.
2. Connect UAV battery negative to J14 pin 2.
3. Verify polarity before applying power.
4. Use a multimeter to confirm expected battery voltage between J14 pin 1 and J14 pin 2.

12.2 Step 2 – BTS7960 Power Wiring

1. Connect J2 pin 2 to BTS7960 motor power positive input.
2. Connect J2 pin 1 to BTS7960 motor power ground.
3. Use short, thick wires for this connection.
4. Keep motor power wiring away from encoder and signal wires where possible.

12.3 Step 3 – BTS7960 Logic Wiring

Table 26: BTS7960 Logic Harness Wiring

Nemo J3 Pin	BTS7960 Pin	Purpose
Pin 1	VCC	Logic 5 V.
Pin 2	GND	Logic ground.
Pin 3	R_EN	Forward enable.
Pin 4	L_EN	Reverse enable.
Pin 5	R_PWM	Forward PWM.
Pin 6	L_PWM	Reverse PWM.
Pin 7	R_IS	Leave unconnected unless current sense is added.
Pin 8	L_IS	Leave unconnected unless current sense is added.

12.4 Step 4 – Pixhawk Wiring

1. Connect Pixhawk telemetry TX to ESP32-C3 RX on J1.
2. Connect Pixhawk telemetry RX to ESP32-C3 TX on J1.
3. Connect Pixhawk ground to system ground.
4. Connect AUX 1 PWM to J7 pin 1 for winch command.
5. Connect AUX 2 PWM to J7 pin 2 for clutch command.

12.5 Step 5 – Encoder Wiring

1. Connect AS5600 VCC to J9 pin 1.
2. Connect AS5600 SCL to J9 pin 2.
3. Connect AS5600 SDA to J9 pin 3.
4. Connect AS5600 GND to J9 pin 4.
5. Keep encoder cable short and routed away from motor wires.

12.6 Step 6 – Servo Wiring

1. Connect servo signal wire to J10 pin 1.
2. Connect servo positive wire to J10 pin 2.
3. Connect servo ground wire to J10 pin 3.
4. Install J12 jumper according to servo voltage requirement.

12.7 Step 7 – Manual Switch and End-Stop

1. Connect end-stop switch to J4.
2. Connect manual up/down switch to J5.
3. Verify switch logic using firmware serial monitor before connecting motor power.

13 Pre-Power Checklist

Before applying power to the board, complete the following checklist.

Table 27: Pre-Power Electrical Checklist

No.	Check Item	Status
1	J14 polarity verified using multimeter.	
2	No short between VBAT_IN and GND.	
3	No short between +5V and GND.	
4	No short between +3.3V and GND.	
5	BTS7960 power wiring polarity verified.	
6	BTS7960 logic wiring order verified.	
7	Servo jumper J12 correctly installed.	
8	External BEC voltage measured before connecting servo.	
9	Pixhawk TX/RX cross-wiring verified.	
10	Encoder I2C wiring verified.	
11	End-stop switch tested manually.	
12	Manual switch neutral position verified.	

14 Power-On Test Procedure

1. Power the board without connecting the motor to the BTS7960 output.
2. Measure the onboard +5V rail.
3. Measure the +3.3V rail.
4. Confirm ESP32-C3 boots correctly.
5. Confirm I2C scan detects AS5600 encoder.
6. Confirm manual switch inputs are detected correctly.
7. Confirm end-stop switch changes state correctly.
8. Confirm servo moves to safe position.
9. Confirm BTS7960 enable lines remain disabled during boot.
10. Connect motor only after logic and safety tests pass.

15 Failsafe and Safety Logic

The firmware should implement multiple safety layers.

15.1 Recommended Failsafe States

Table 28: Recommended Failsafe Behavior

Fault Condition	Required Action
Pixhawk signal lost	Disable motor PWM and hold clutch in safe state.
UART telemetry lost	Continue local safety control and report fault when communication returns.
End-stop triggered during retract	Immediately stop motor and block further retract command.
Both manual switch directions active	Disable motor. Treat as invalid input.
Encoder not detected	Disable automatic depth control. Allow only manual low-speed recovery if safe.
Brownout detected	Disable BTS7960 enable lines and reset output states.
Servo BEC missing	Prevent clutch movement command and report servo power fault.

15.2 Boot Default States

At boot, the following default states are recommended:

- `R_PWM = LOW`
- `L_PWM = LOW`
- `R_EN = LOW`
- `L_EN = LOW`
- Servo moves to predefined safe position after power stabilization.
- Encoder is initialized before motor motion is allowed.

16 Ground Operator Quick Reference

Table 29: Operator Quick Reference

Operation	Procedure
Power On	Connect UAV payload battery to J14. Verify status LEDs and telemetry.
Manual Spool Up	Move manual switch to UP position. Release switch to stop.
Manual Spool Down	Move manual switch to DOWN position. Release switch to stop.
Zero Encoder	Activate ZERO RESET input after the spool is placed at known home position.
Servo Power Selection	Use J12 pins 2–3 for internal 5 V servo, or pins 1–2 for external BEC servo.
Emergency Stop	Remove motor enable command, release manual switch, or disconnect main power if required.

17 Recommended Harness Labeling

Each harness should be labelled with heat-shrink or printed tags.

Table 30: Harness Labeling Recommendation

Harness	Recommended Label
Battery Input	MAIN VBAT IN – J14
BTS7960 Power	MOTOR DRIVER POWER – J2
BTS7960 Logic	BTS7960 LOGIC – J3
Pixhawk Telemetry	PIXHAWK UART – J1
Pixhawk AUX PWM	WINCH CONTROL INPUT – J7
Encoder	AS5600 ENCODER – J9
Servo	CLUTCH SERVO – J10
End-Stop	RETRACT LIMIT SWITCH – J4
Manual Switch	MANUAL OVERRIDE – J5
External BEC Input	BEC INPUT VBAT – J11
External BEC Output	BEC OUTPUT SERVO POWER – J13

18 Common Assembly Mistakes to Avoid

- Do not power a high-torque servo from the internal 5 V rail.
- Do not install the J12 jumper between pin 1 and pin 3.
- Do not connect BTS7960 motor power to the 5 V logic rail.
- Do not wire UART TX-to-TX and RX-to-RX. Pixhawk TX must go to ESP32 RX.
- Do not route AS5600 I2C wires alongside motor power wires.
- Do not leave the end-stop input floating without pull-up configuration.
- Do not allow both forward and reverse PWM channels to be active together.
- Do not connect the motor before validating firmware output states.
- Do not assume servo wire color order blindly. Always verify signal, VCC, and GND.

19 Final Integration Notes

For flight use, the winch system should be tested in this order:

1. Bench test without motor.
2. Bench test with motor disconnected from spool.
3. Bench test with spool and no payload.
4. Ground test with dummy payload.
5. Tethered UAV test.
6. Low-altitude flight test with dummy payload.
7. Full mission test after safety validation.

The final system must always default to a safe state when command signal, telemetry, or power stability is lost.

End of Document
Nemo Winch V3.0 – Geared DC Motor Iteration

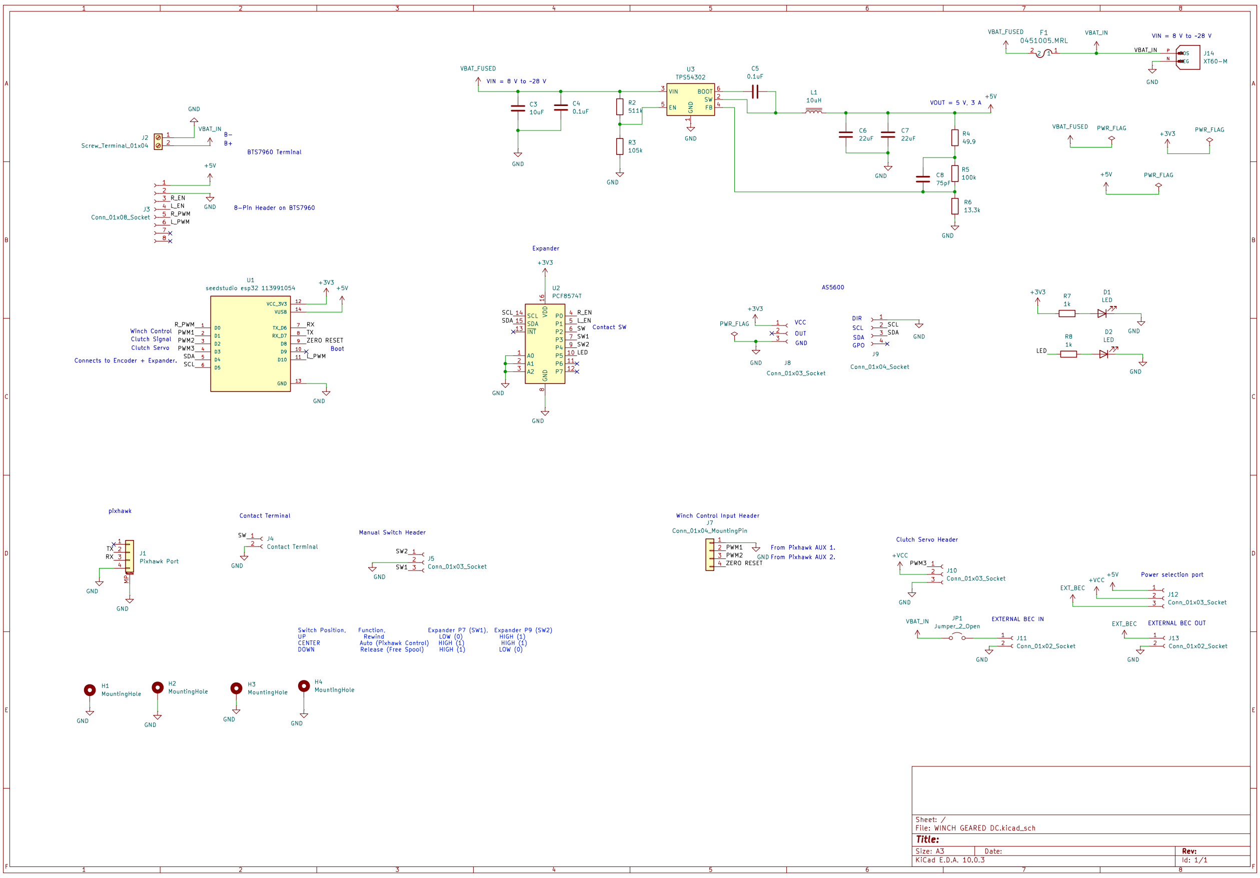

Figure 1: schematic